

TOPTIER-CNS Summer School 2026

(TOPTIER-CNSSS26)

Date_ 23-28 Aug. 2026

Venue_ IBS Science Culture Center, Daejeon



Probing Neutron–Neutron Short-Range Correlations in ^{16}O

The $^{16}\text{O}(p,nd)^{14}\text{O}$ Reaction at RCNP

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Outline

- 1 | Introduction
 - 2 | Experimental procedure
 - 3 | Data analysis
 - 4 | Preliminary results
-

Introduction: Short Range Correlations



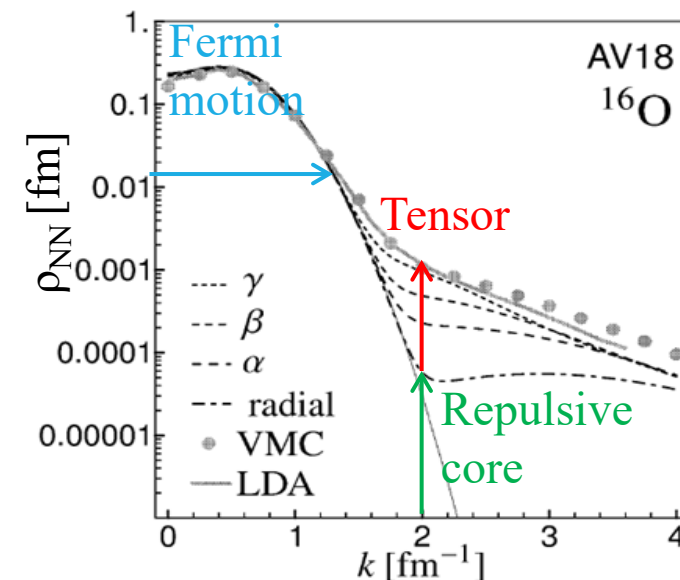
High-momentum components above k_F arise from nucleon-nucleon correlations.

Tensor interaction

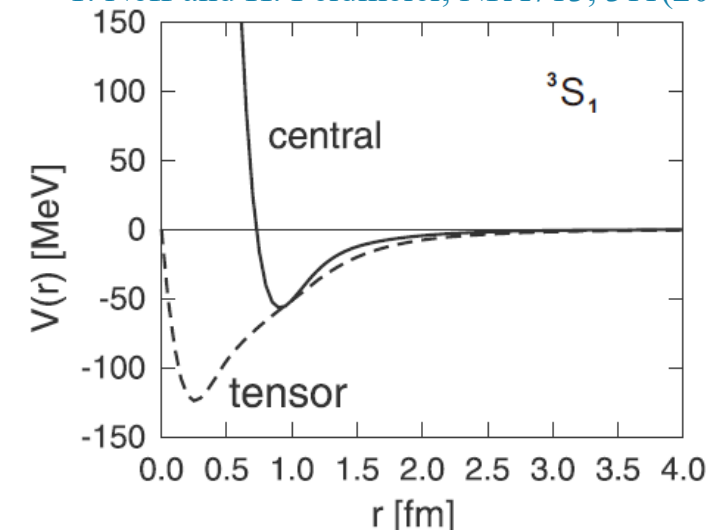
- Originate from π exchange
- Non-central, 70–80% of NN attraction
- Contribute mainly to the $(S, T)=(1, 0)$ channel
- 2p–2h excitation to higher orbitals with $\Delta S=\Delta L=2$

Short-range central interaction

- Central and repulsive
- Approximately symmetric among different ST channels
- Gives rise to a high-momentum tail through hard collisions at short distances

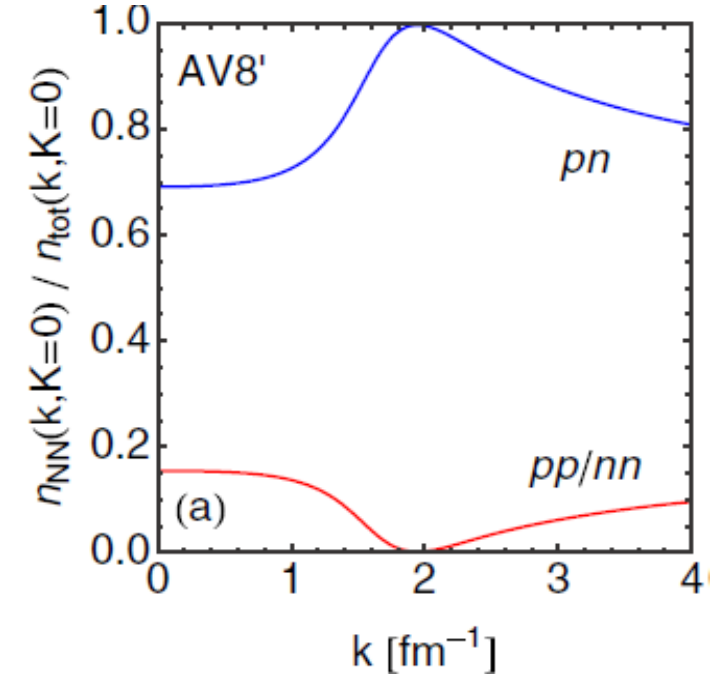
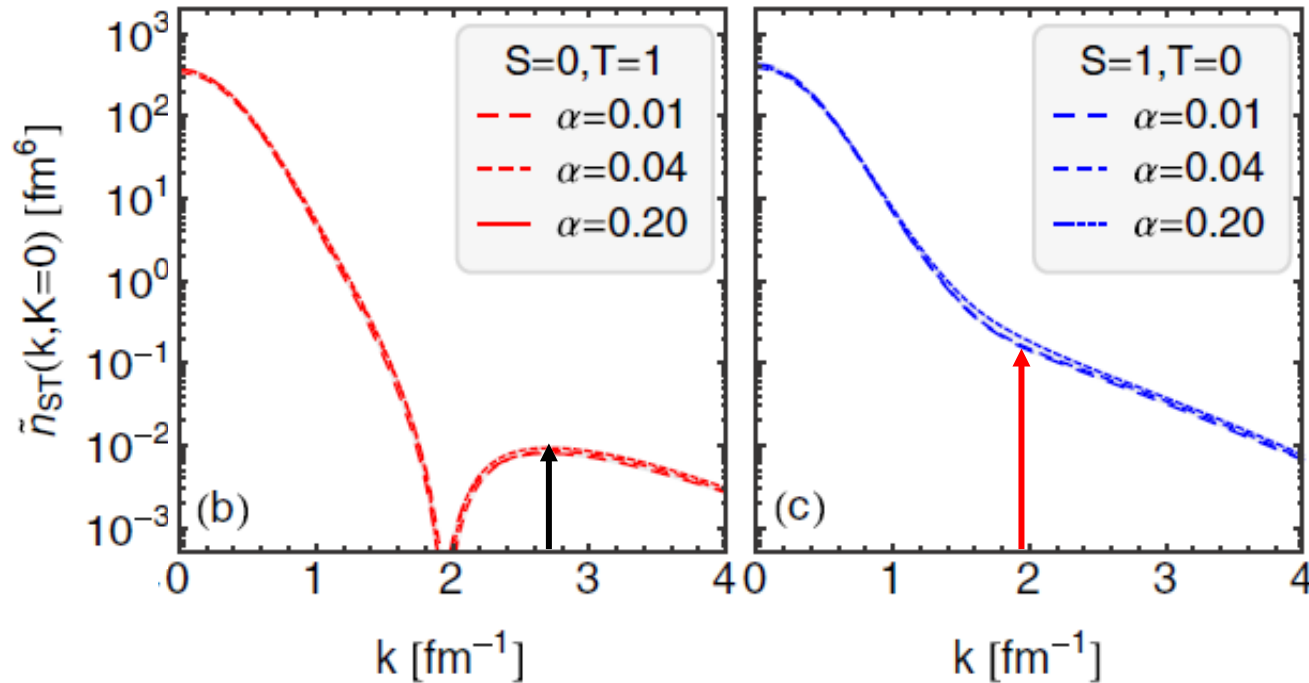


T. Neff and H. Feldmeier, NPA713, 311(2003)



K. Ikeda and H. Toki, Lecture Notes in Physics, Springer, 165 (2010)

Theoretical predictions of momentum distribution



- At around **2 fm⁻¹**, np (S, T)=(1,0) pairs dominate due to the strong tensor interaction.
- Short-range repulsion generates a high-momentum tail extending to very high k .

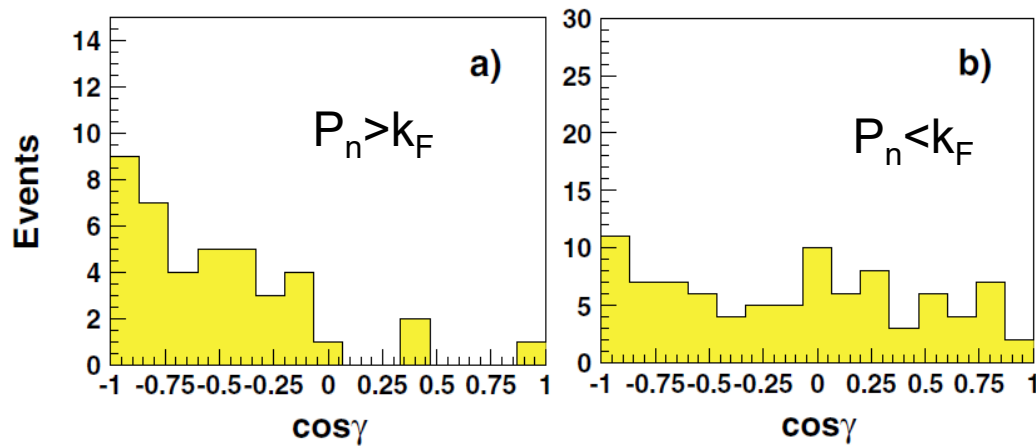
- Relative probability to find pn or pp/nn pairs is a function of relative momentum.
- Provides an opportunity to investigate the short range correlations

T.Neff et al., PRC 92(2015)024003

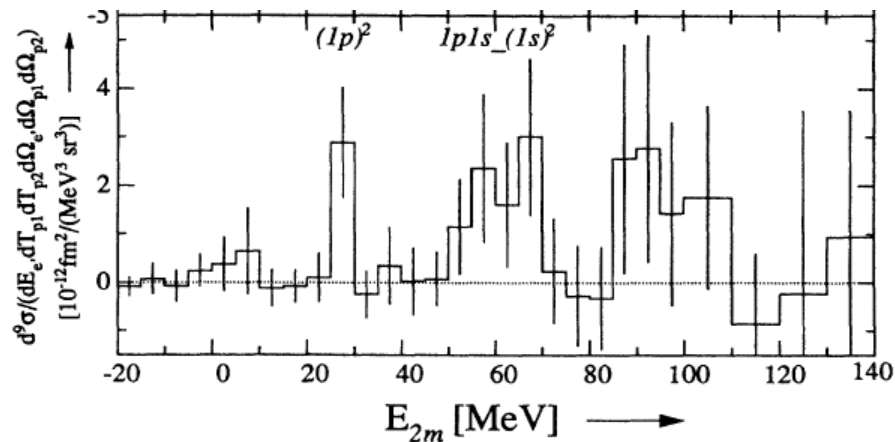
Previous experimental studies

Two-nucleon knockout reaction

$^{12}\text{C}(p,2p+n)$ @ BNL A.Tang et al., PRL 90 (2003) 042301

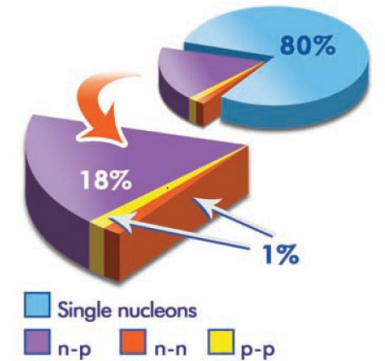
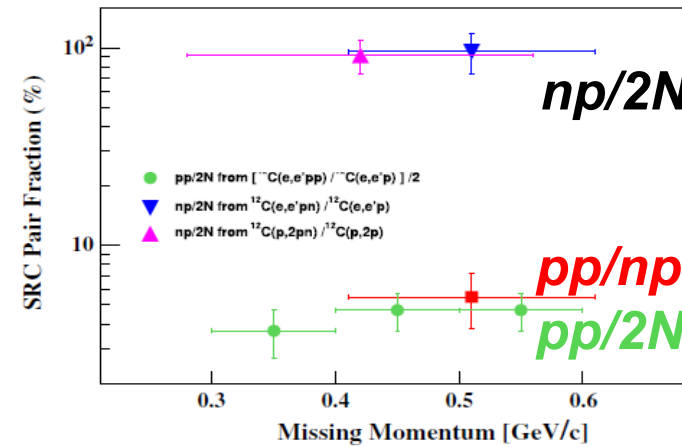


$^{12}\text{C}(e,e'pp)$ @ NIKHEF L.J.H.M.Kester et al., PRL 74(1995) 1712



$^{12}\text{C}(e,e'pN)$ @ JLab

R.Subedi et al., Science 320 (2008) 1476



- Strong suppression of pp pairs is observed compared with np pairs, with an np/pp ratio of about **18**.
- These measurements provide limited information on the **spin-isospin dependence** of the correlated pairs.
- More statistics and moderate resolution are needed for further investigation.

(p,Nd) reaction is a possible method

Experimental procedures: setup

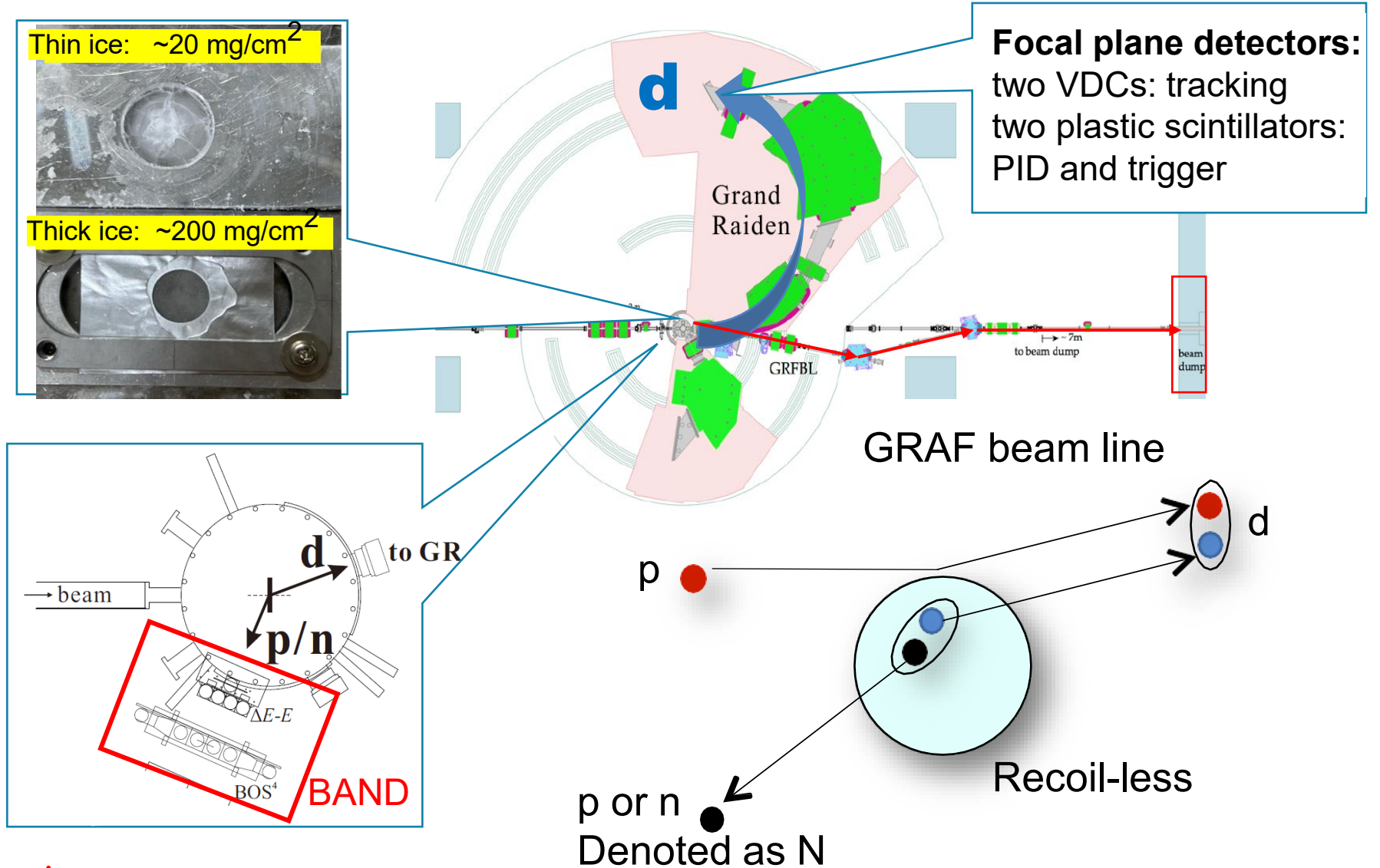
E552@WS course, RCNP

- Beam: 392, 230 MeV
- Target:
 - thin ice: $^{16}\text{O}(p, \textcolor{red}{p}d)^{14}\text{O}$
 - thick ice: $^{16}\text{O}(p, \textcolor{red}{n}d)^{14}\text{O}$

BAckward **N**ucleon **D**etectors (BAND)

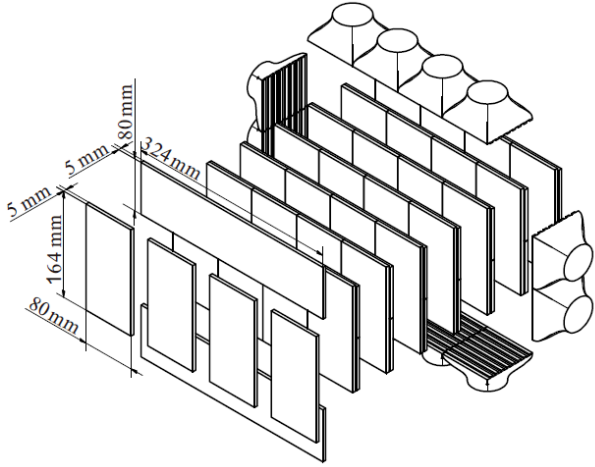
- ΔE -E: proton detector
- BOS⁴: neutron detector

Low-background coincidence measurement

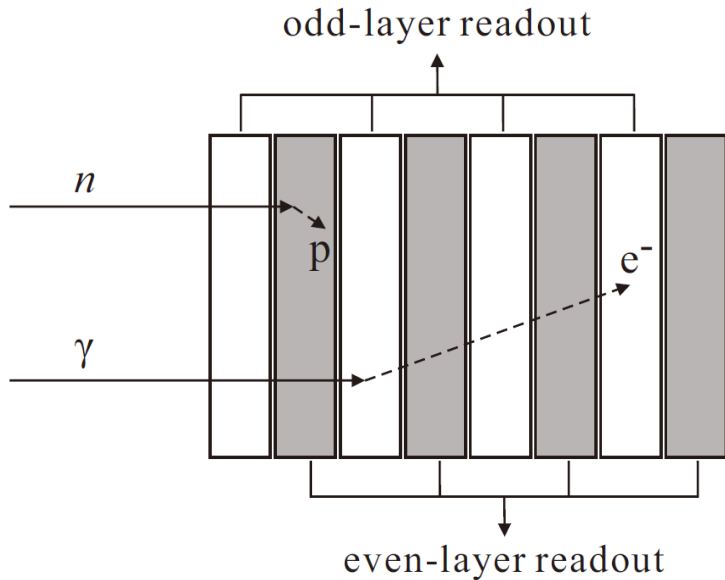


Experimental procedures: BOS⁴ detector

Structure and Discrimination Principle



Beihang-Osaka university
Stack Structure Solid
Organic Scintillator
(BOS4)



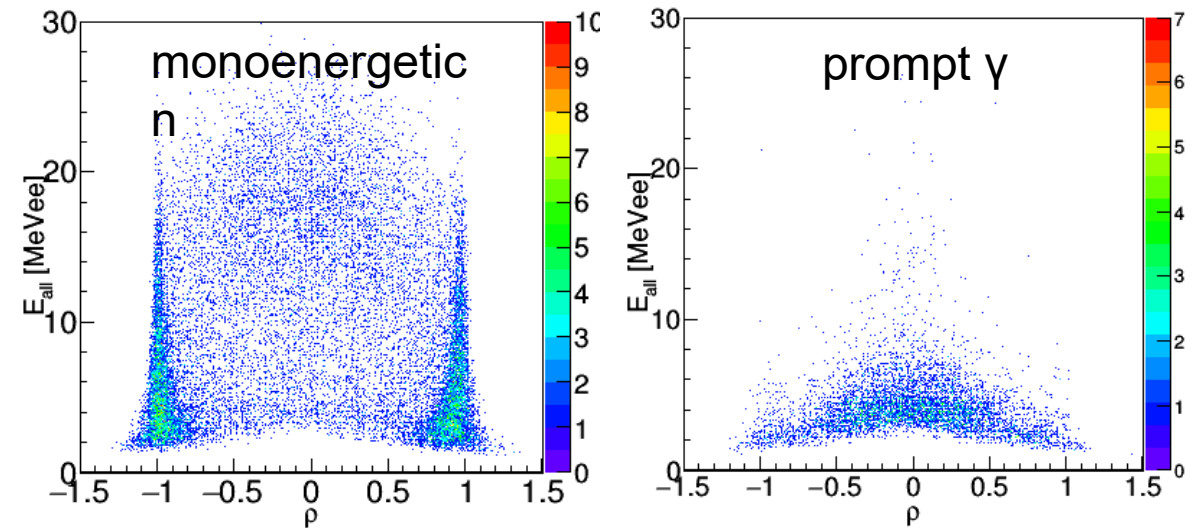
$$E_{\text{all}} = E_{\text{odd}} + E_{\text{even}}$$

$$\rho = \frac{E_{\text{even}} - E_{\text{odd}}}{E_{\text{even}} + E_{\text{odd}}}$$

L. Yu et al, NIM A 866(2017)118

n- γ discrimination under a high counting rate Experimental results

- Monoenergetic neutrons at 31 MeV produced by the $d+d \rightarrow n+{}^3\text{He}$ reaction

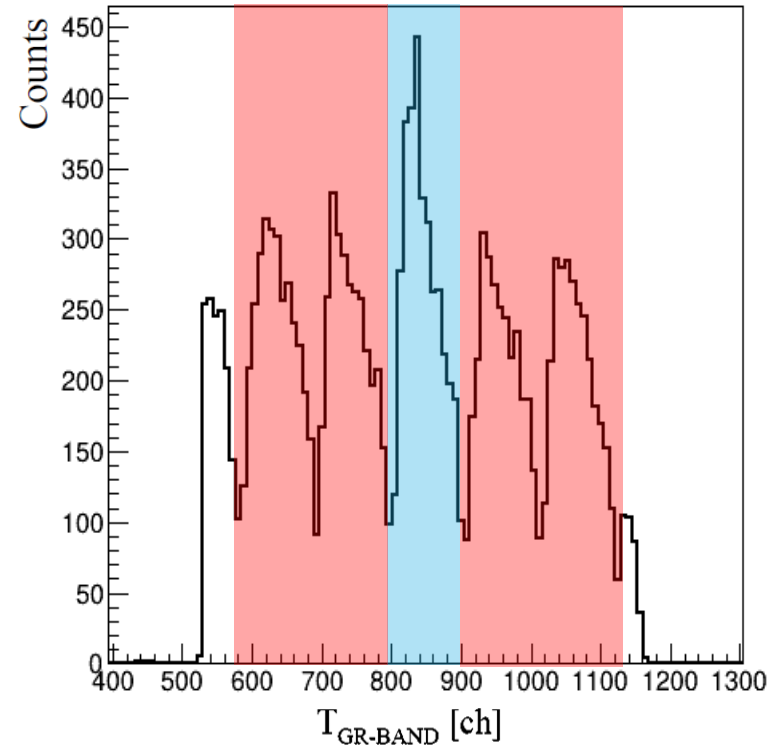


neutron: $\rho \sim \pm 1$

γ ray: $\rho \sim 0$

Clear n- γ separation in E_{all} - ρ plot

Data analysis: coincidence measurement of (p,nd) reaction



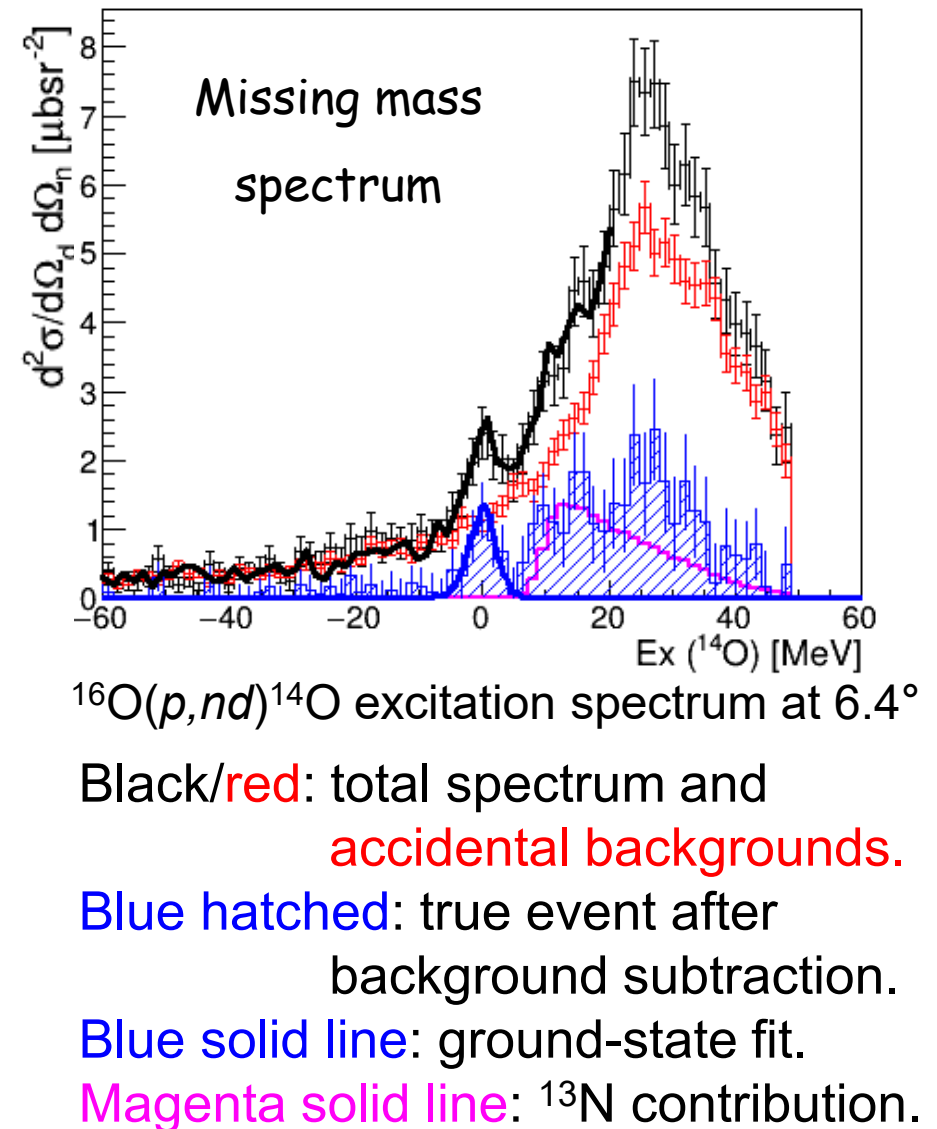
← Time difference between deuteron and neutron triggers

- Prompt peak: true + accidental coincidence
- Other 4 peaks: accidental coincidence between different beam bunches

True: Acc. coin: ~ 1:5

After subtracting accidental coincidence,

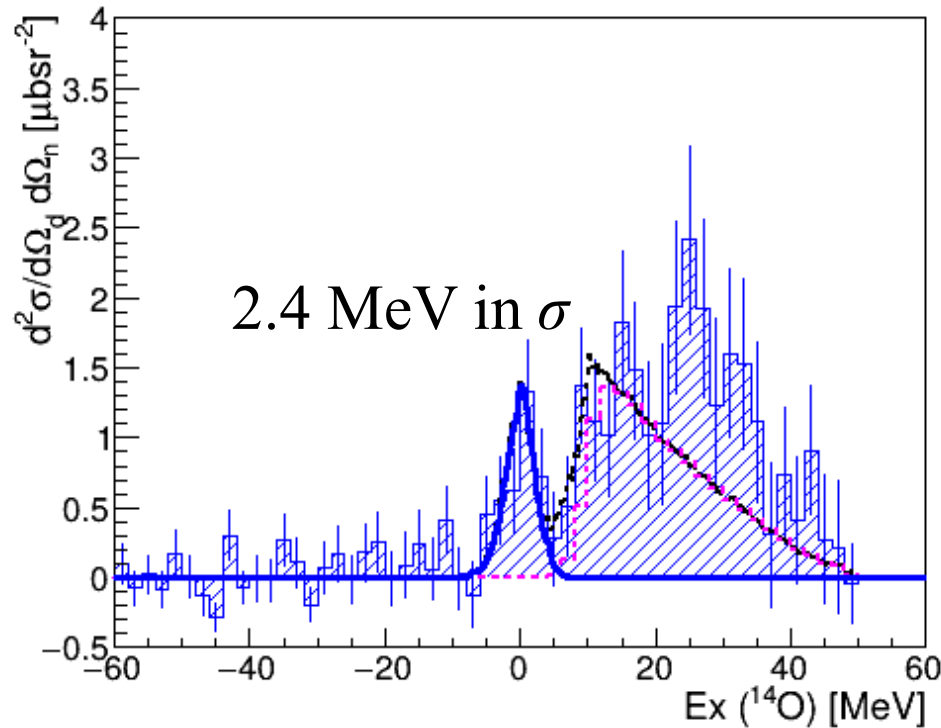
G.S. ($0^+, 1$) can be clearly observed.



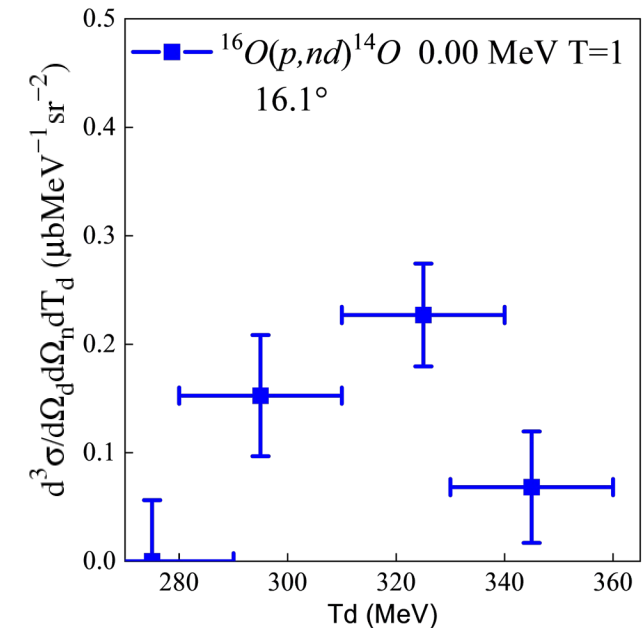
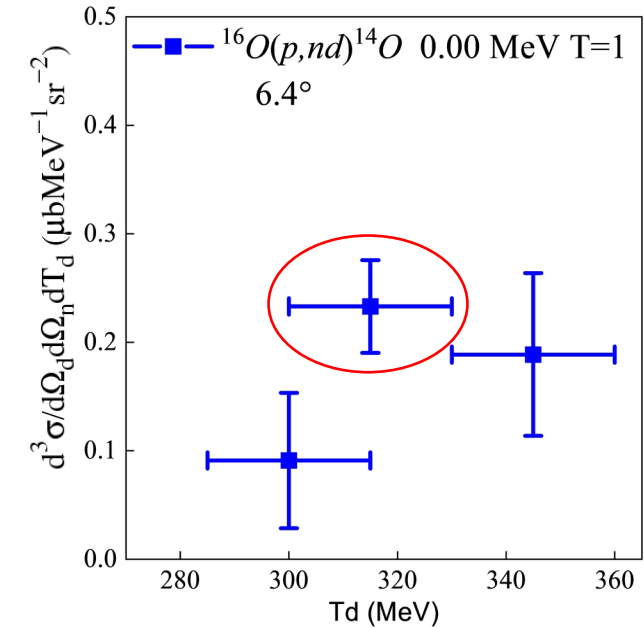
Data analysis: (p,nd) triple differential cross section

The (p,nd) triple differential cross section along the E_x axis is given by:

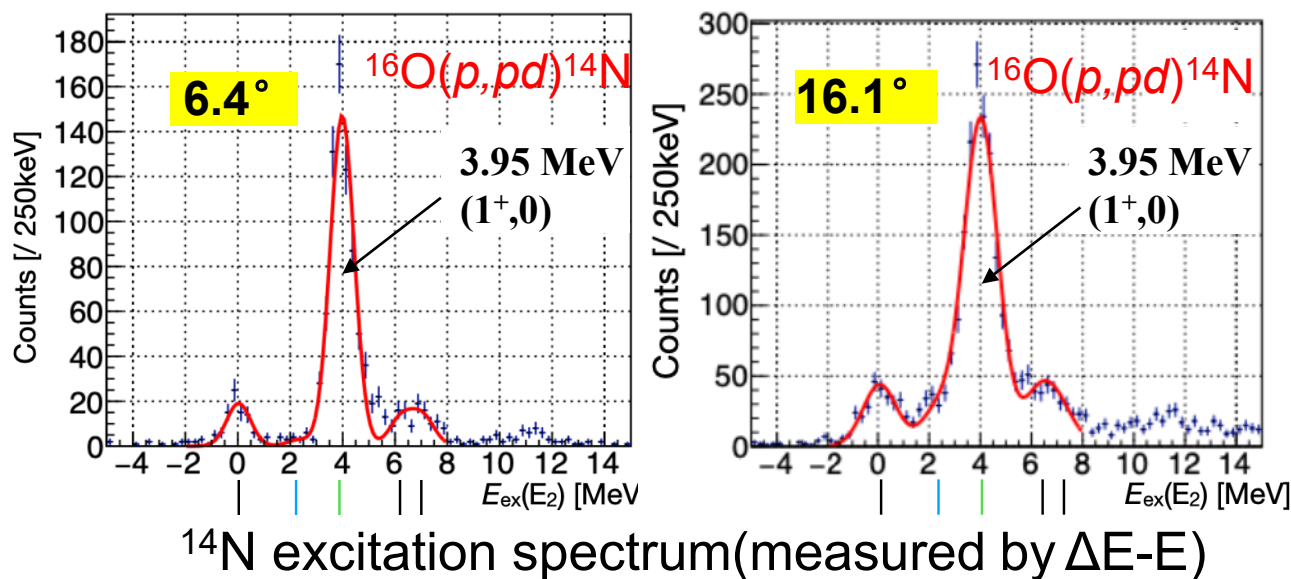
$$\frac{d^3\sigma}{d\Omega_d d\Omega_n dT_d} = \frac{N_{(p,nd)}^{g.s.}}{N_t N_b \epsilon_{DAQ} \epsilon_d \epsilon_n} \cdot \frac{1}{\Delta\Omega_d \Delta\Omega_n \Delta T_d},$$



$^{16}\text{O}(p,nd)^{14}\text{O}$ excitation spectrum at 6.4° , Bp ratio = 1.21



Preliminary Results: compare with (p,pd) channel



X. Wang, Ph.D. thesis, 2025.

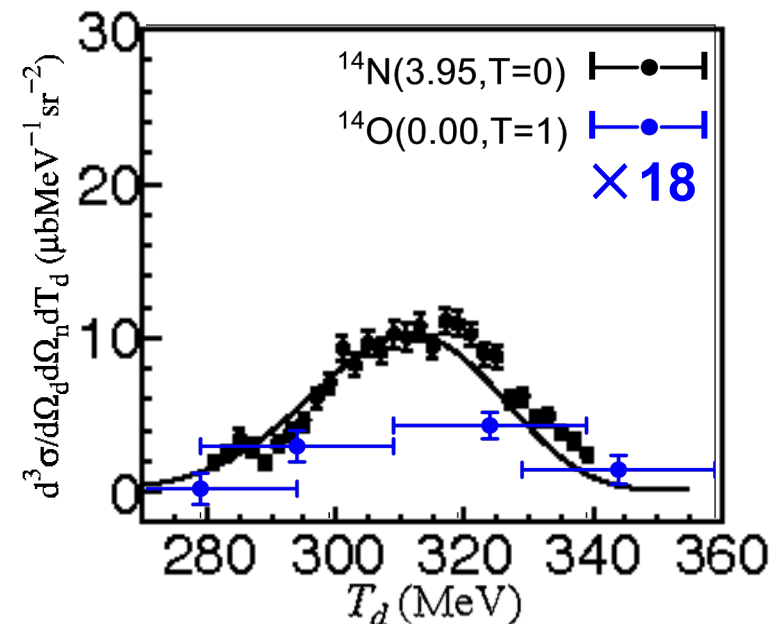
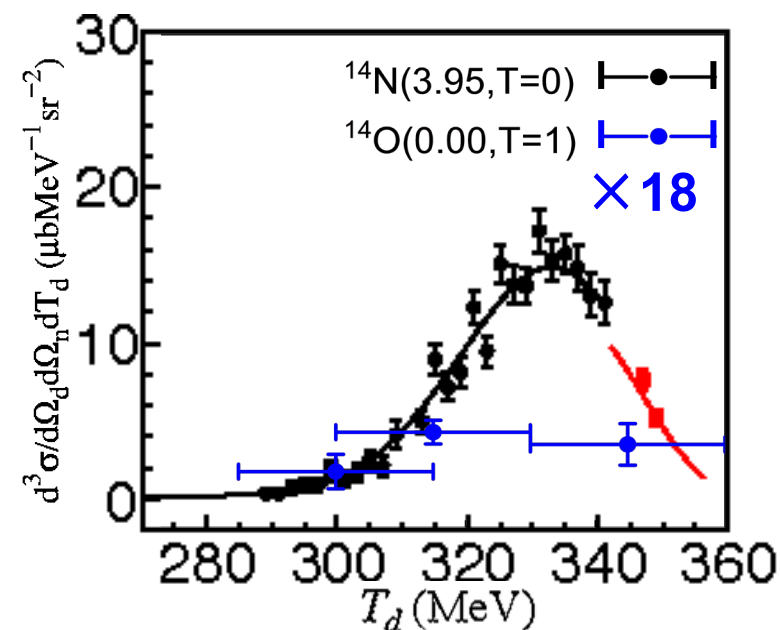
$^{16}\text{O}(p,pd)^{14}\text{N}$ 3.95 MeV (1^+ , $L=0$)

$^{16}\text{O}(p,nd)^{14}\text{O}$ 0.00 MeV (0^+ , $L=0$)

The np/nn ratio of the triple differential cross section is about 18.

Dominance of np ($ST10$) pair over nn ($ST01$) pair.

Direct evidence of tensor effect!



- The n-n pair is observed at around 2 fm^{-1} via $^{16}\text{O}(p,nd)^{14}\text{O}$ reaction
 - The ground state of ^{14}O was identified, and the triple differential cross section for the ground-state transition was extracted.
 - The nn yield is found to be strongly **suppressed** compared with the pn case, suggesting a possible signature of the dominant tensor force effect at $\sim 2 \text{ fm}^{-1}$
- The (p,Nd) reaction proves to be a promising probe of high-momentum components in nuclei and can be extended to other nuclei beyond ^{16}O

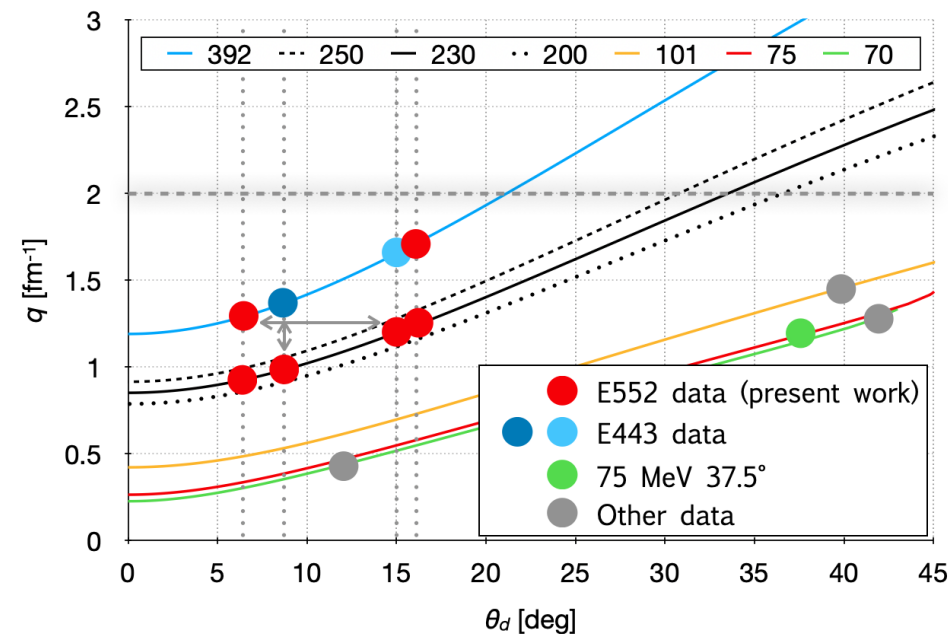
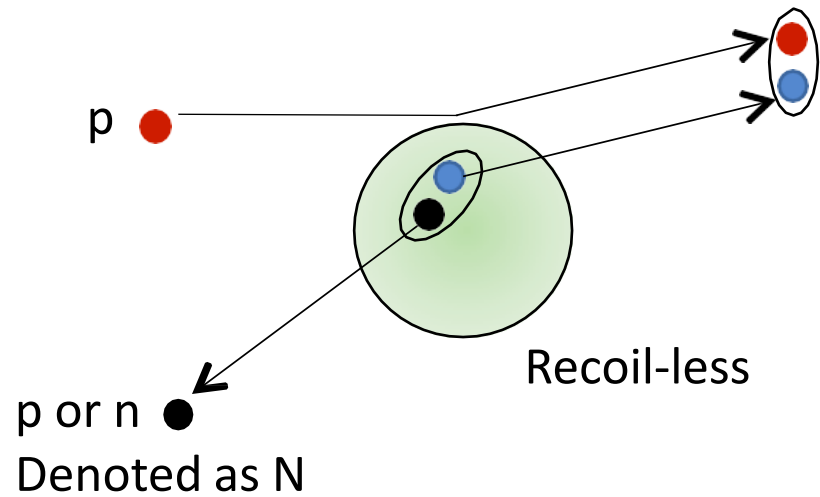
Thank you for your attention!

Backup

kinematics settings



- Measurements were performed at beam energies of 392 MeV (two angles) and 230 MeV (four angles), with several momentum settings
- covering 0.9–1.8 fm⁻¹ of momentum between pickup neutron and eject neutron/proton

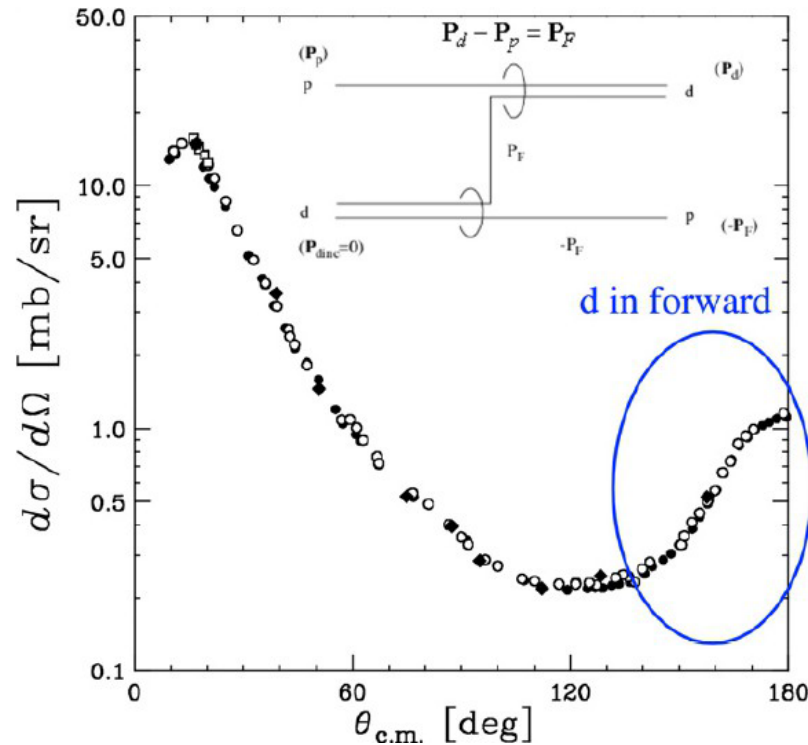


Momentum transfer of ¹⁶O(*p*,*Nd*) reaction

T_b [MeV]	θ_d	q [fm ⁻¹]	Proton angle(HODO)	Neutron angle(BOS ⁴)
392	6.4°	1.29	132.8° – 158.7°	142.5° – 155.9°
392	16.1°	1.71	107.5° – 138.8°	106.4° – 119.8°
230	6.4°	0.94	132.8° – 158.7°	142.5° – 155.9°
230	8.7°	1.00	123.5° – 149.4°	133.3° – 146.7°
230	15.0°	1.22	110.3° – 136.7°	109.3° – 122.7°
230	16.1°	1.26	107.5° – 138.8°	106.4° – 119.8°

Kinematics summary

(p,d) Reaction at high momentum transfer



K.Sekiguchi et al., PRL 95 (2005) 162301

Forward d in (p,d) reaction is equivalent to backward elastic scattering of $p+d$, which is dominated by a neutron pick-up

$$\sigma_F = K \frac{P_d}{P} N(P_F) \left[B_D + \frac{\hbar^2}{M} (\mathbf{p} - \mathbf{P}_d/2)^2 \right]^2 \left| \langle \varphi(r), e^{i(\mathbf{p} - \mathbf{P}_d/2) \cdot \mathbf{r}} \rangle \right|^2$$

K: phase space constant, B_D : deuteron binding energy, M: nucleon mass by G. F Chew and M.L. Goldberger Phys. Rev. 77 (1950) 470.

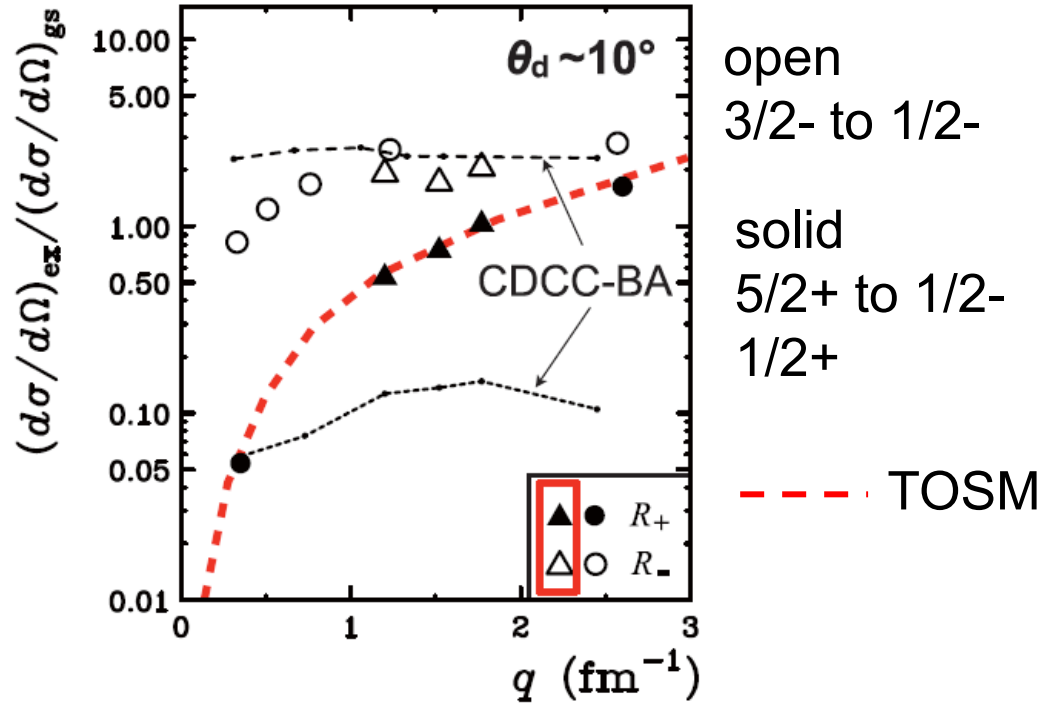
Reaction at “backward” occurs via the pick-up of high-momentum neutrons.

Momentum selectivity of neutron pickup reaction: momentum transferred to the deuteron = initial momentum of picked-up neutron.

High-momentum n - N pairs can be probed by the (p, Nd) reaction!

(p,d) Reaction at high momentum transfer

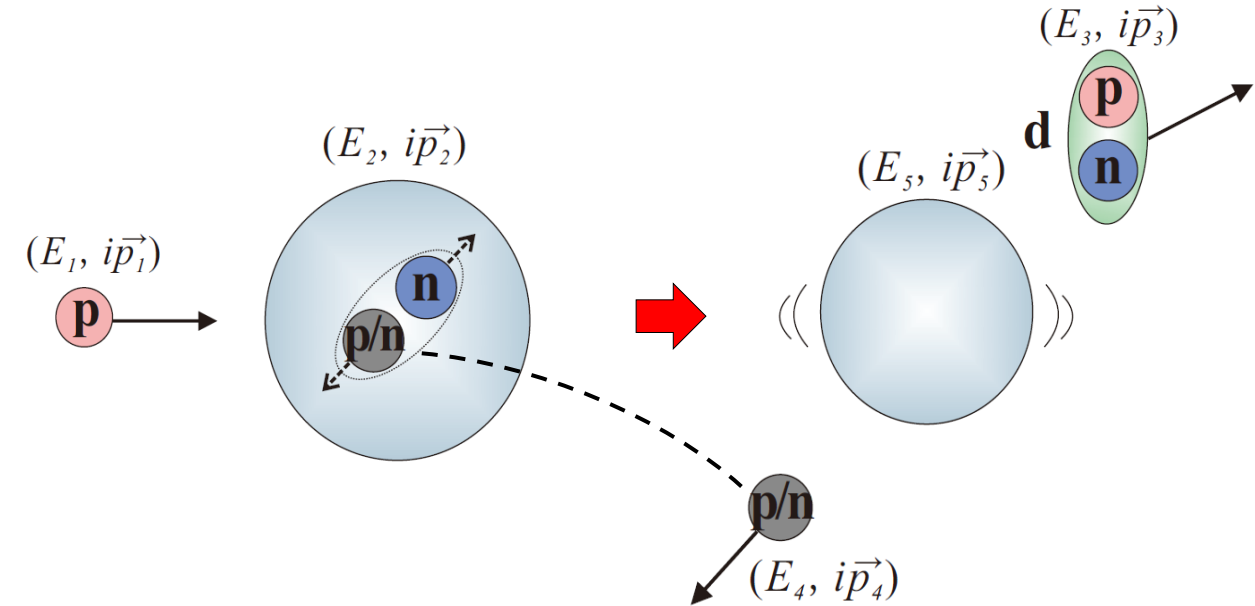
- $^{16}\text{O}(p,d)$ at 198, 295, 392 MeV @RCNP



H.J.Ong et al, PLB 725(2013)277

The (p,d) reaction is sensitive to tensor interactions and high-momentum components, particularly at large momentum transfer

? What kind of correlation induces the high-momentum neutrons in (p,d) pickup reaction?



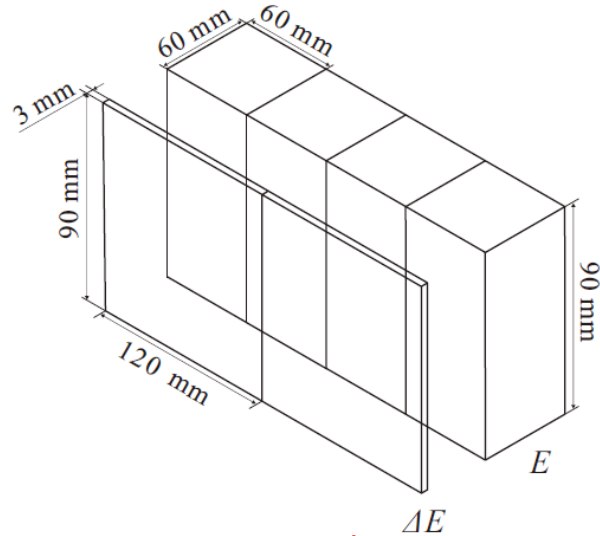
residual nucleus and spin-isospin state:

$(1^+, 0)$ state of ^{14}N $\leftrightarrow$ n-p (ST10)
 $(0^+, 1)$ state of ^{14}N $\leftrightarrow$ n-p (ST01)
 $(0^+, 1)$ state of ^{14}O $\leftrightarrow$ n-n (ST01)

BACkward Nucleon Detectors (BAND)

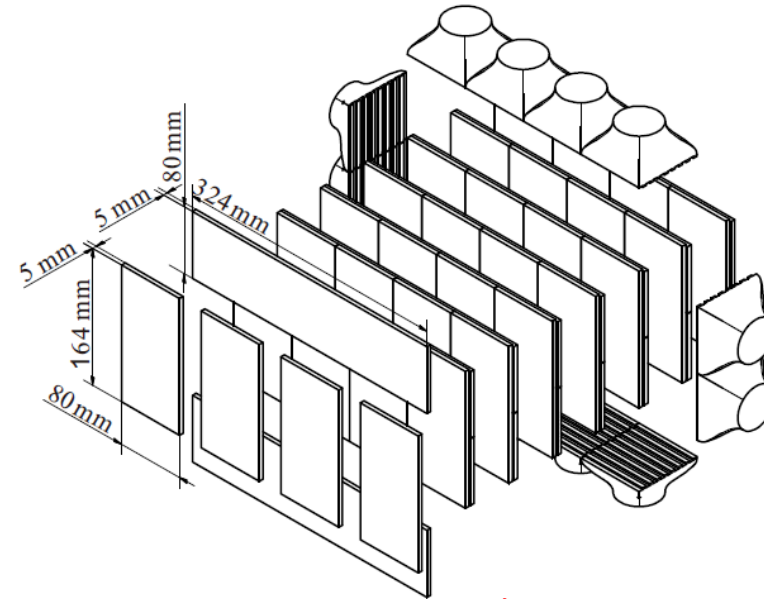


ΔE -E + BOS⁴



proton detection

- Name: ΔE -E telescope
- Type: BC408
- Volume: $(3+60)t \times 240 \times 90 \text{ mm}^3$
- Flight path (middle): $\sim 46\text{cm}(\Delta E)/52\text{cm}(E)$



neutron detection

Multi-layer plastic scintillation detector (BOS4)

- Type: BC408
- Volume: $(5 \times 16)t \times 320 \times 160 \text{ mm}^3$
- Flight path: $\sim 136 \text{ cm}$

Structure of BOS⁴ Detector



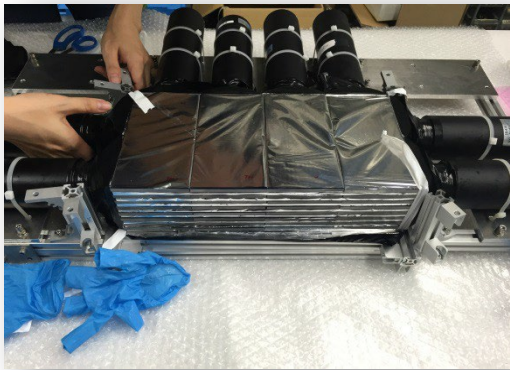
--Configuration

Area: $320 \times 160 \text{ mm}^2$, total thickness: 80mm

16 layers of plastic scintillator (BC-408)

even(H): $5^t \times 320 \times 80 \text{ mm}^3 \times 2 \times 8$ layers

odd(V): $5^t \times 160 \times 80 \text{ mm}^3 \times 4 \times 8$ layers



--Reflector

All plates are wrapped by 12- μm Aluminized Mylar film

--Light guide x 12

Sawtooth shape to connect odd or even layers only

--Photomultiplier x 12

H7195/H6410



--Performance

Counting rate: $>50 \text{ kHz}@20 \text{ nA}$

Position resolution: 2.4 cm in σ Timing

resolution: 0.28 ns in σ

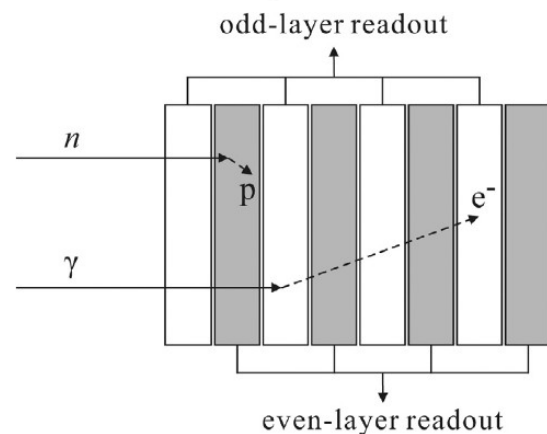
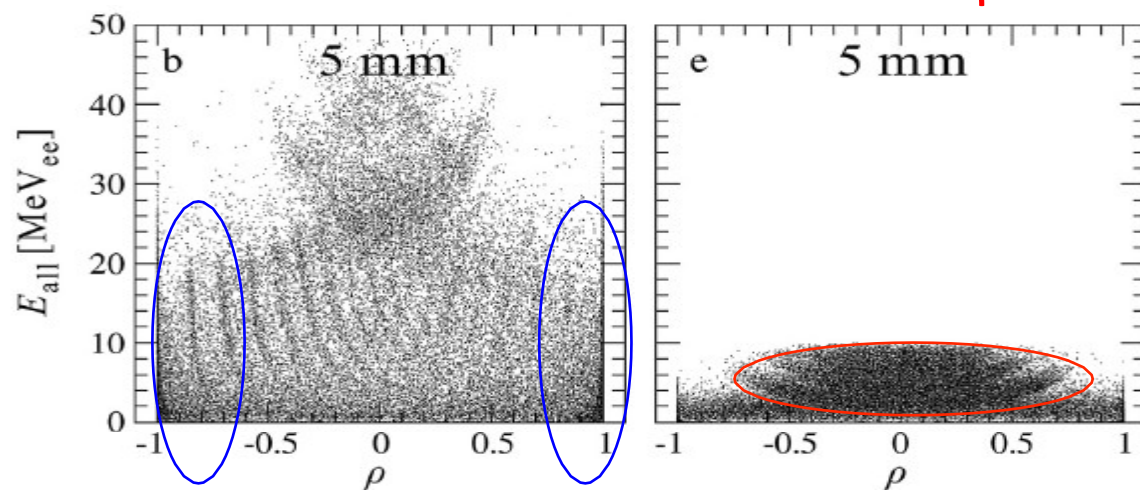
Light output resolution: $\sigma(L)/L = 0.33/\sqrt{L}$

Capability of n-γ discrimination using range + TOF under high-counting rate

- Discrimination principle:

Simulation by Geant4

20~60 MeV n 0~10 MeV γ



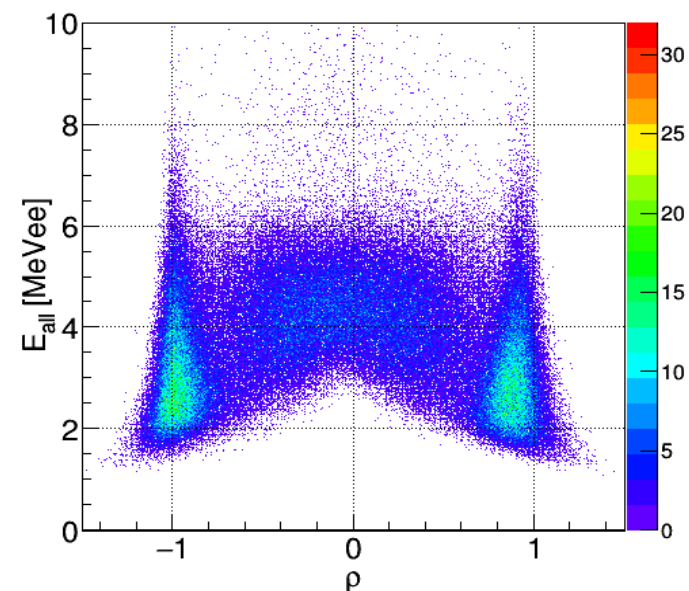
$$E_{\text{all}} = E_{\text{odd}} + E_{\text{even}}$$

$$\rho = \frac{E_{\text{even}} - E_{\text{odd}}}{E_{\text{even}} + E_{\text{odd}}}$$

- Detector configuration

- 5-mm-thick normal plastic scintillator (BC-408) x 16 layer
- Total area: 320 x 160mm²
- Odd and even layers are read out separately.

Test by ²⁴¹Am-⁹Be



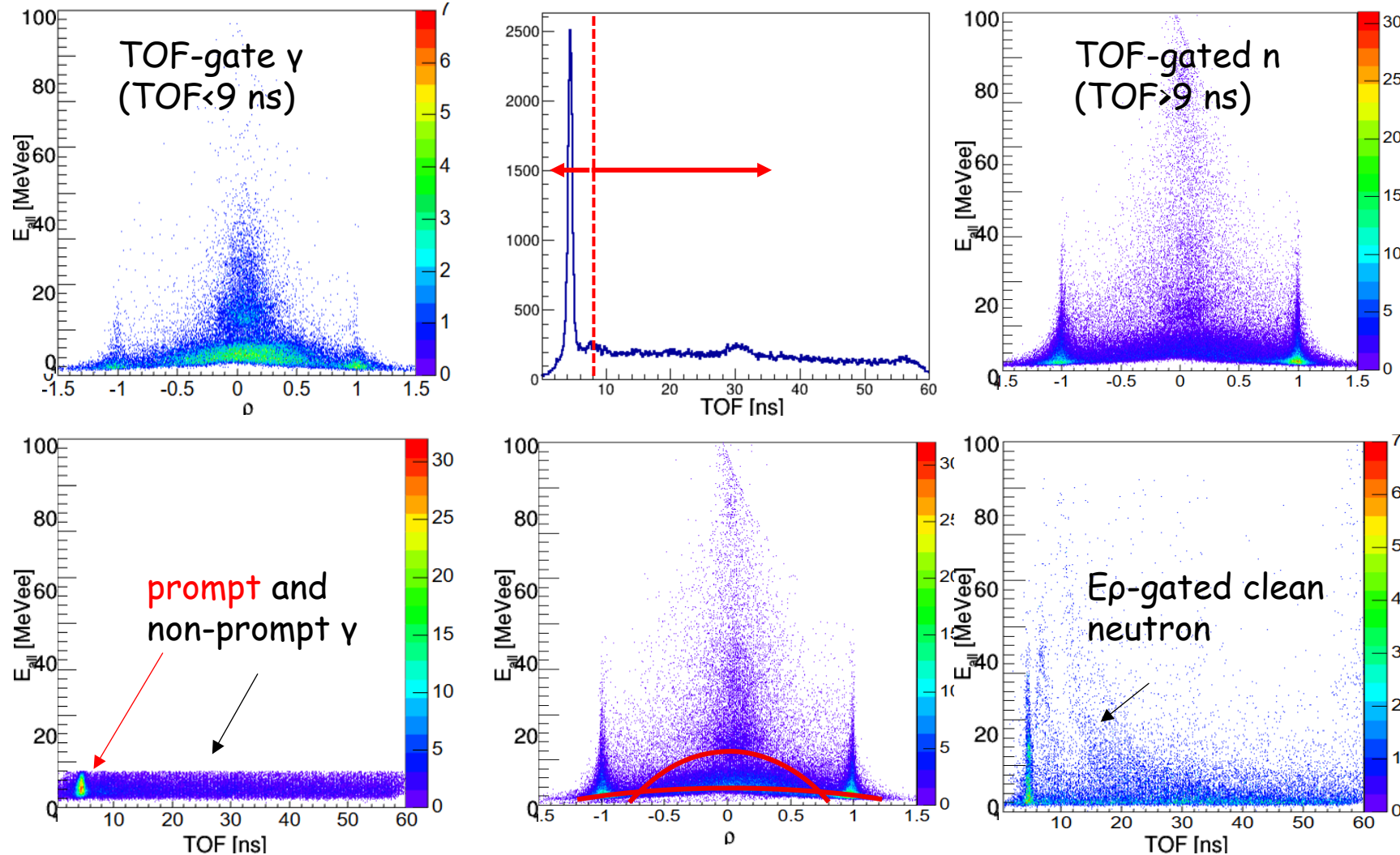
L. Yu et al, NIM A 866(2017)118-128

Data analysis: Neutron analysis



Neutron identification

Charge veto + TOF gate + E- ρ gate

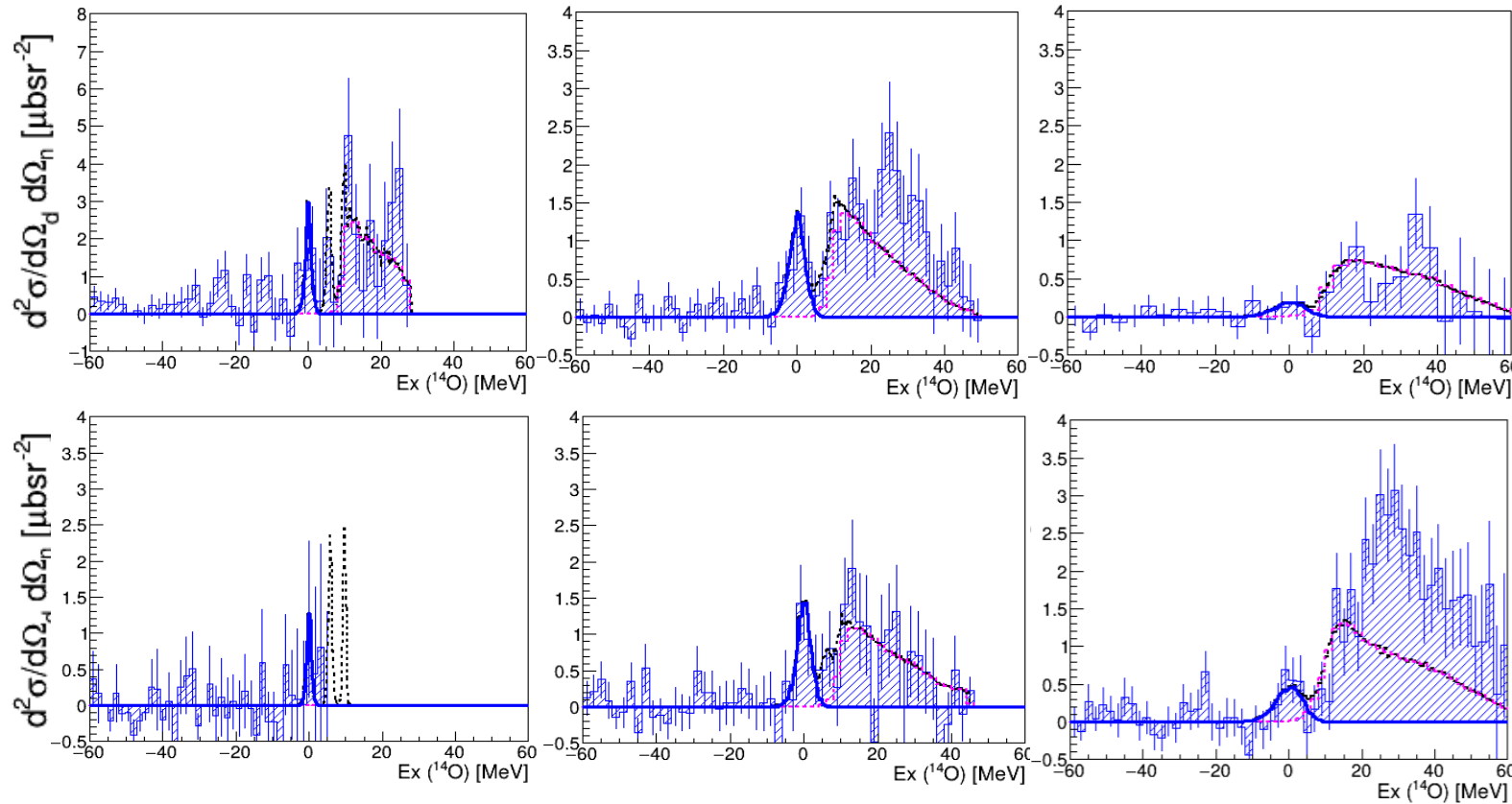


- Neutrons are clearly identified using E_{ρ} +TOF information.

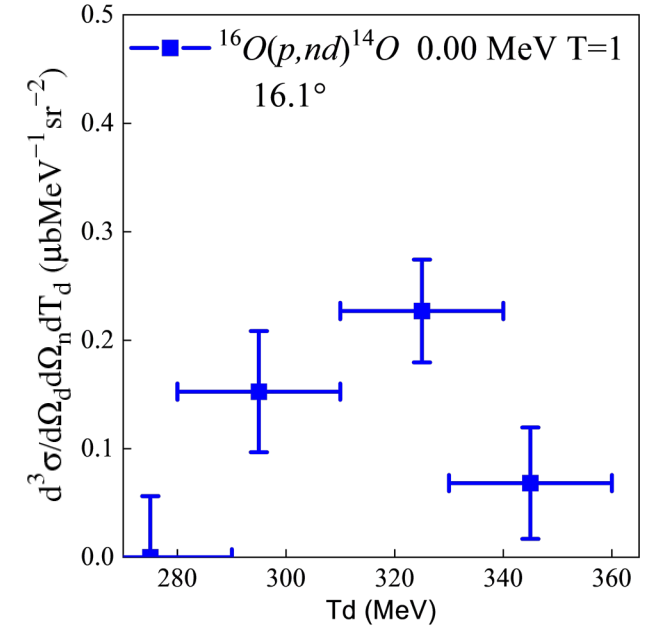
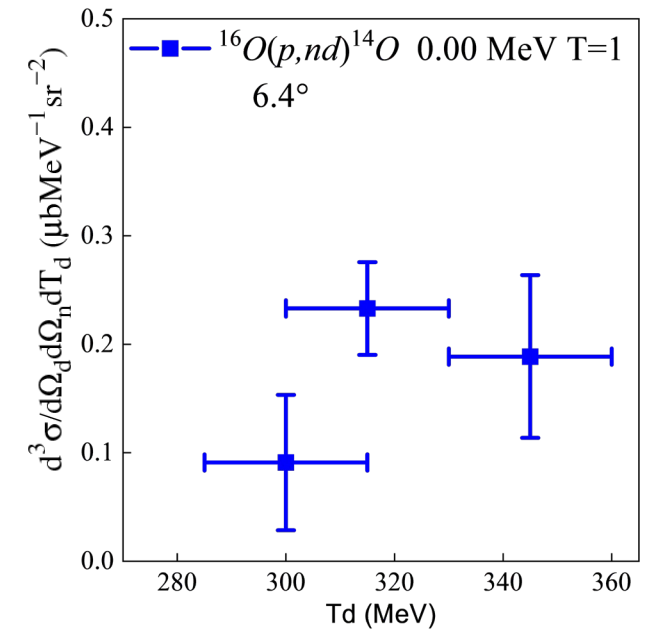
- **TOF and position** on BOS4 reconstruct the momentum of neutrons.

Data analysis: (p,nd) triple differential cross section

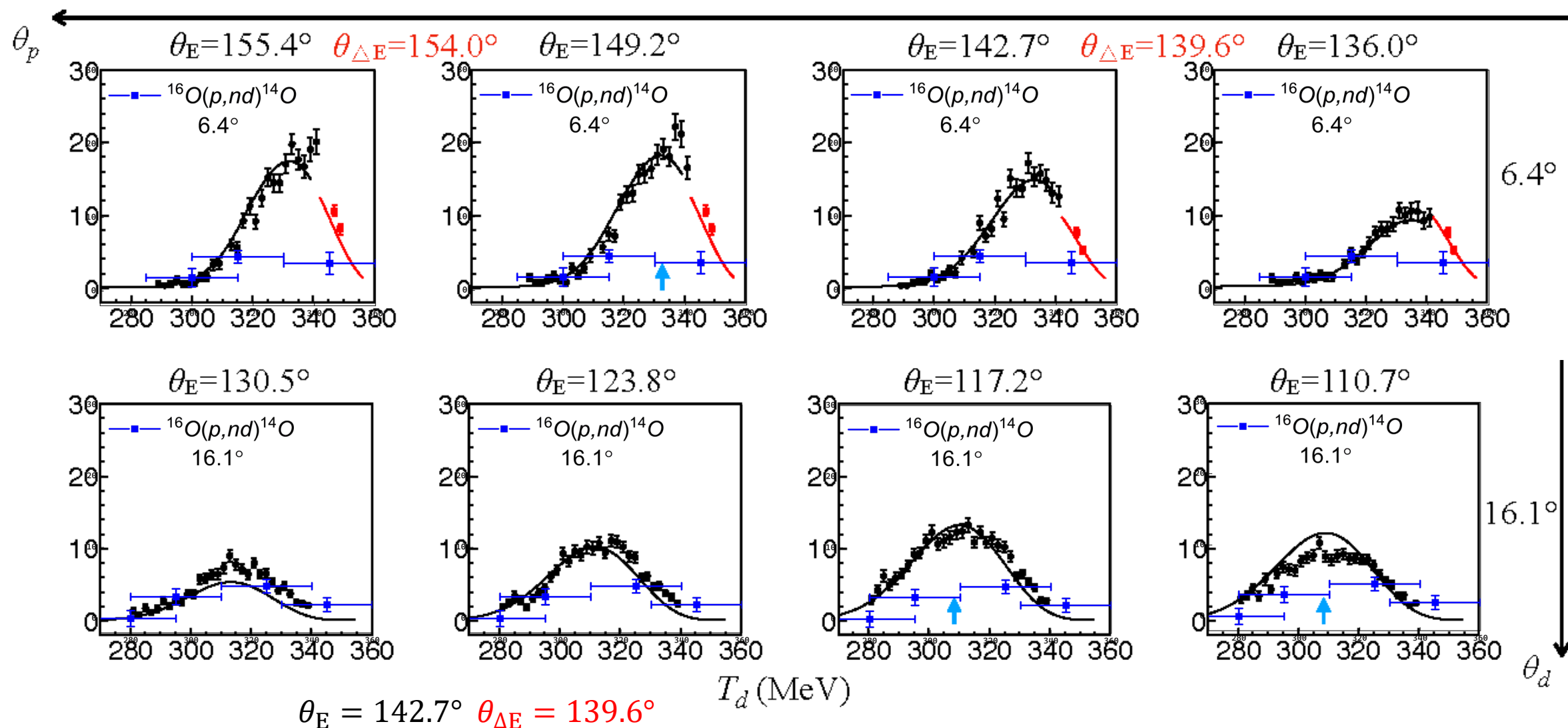
$$\frac{d^3\sigma}{d\Omega_d d\Omega_n dT_d} = \frac{N_{(p,nd)}^{g.s.}}{N_t N_b \epsilon_{DAQ} \epsilon_d \epsilon_n} \cdot \frac{1}{\Delta\Omega_d \Delta\Omega_n \Delta T_d},$$



$^{16}\text{O}(p,nd)^{14}\text{O}$ excitation spectrum at 6.4°(upper), 16.1°(lower)



Preliminary Results: compare with (p,pd) channel



Triple differential cross sections of $^{16}\text{O}(p,nd)$ E443



E443 392 MeV @RCNP

Averaged cross-section of G.S.
in four settings
(extracted from fitting)

Comparison with $^{16}\text{O}(p,pd)$ channel
(measured by ΔE -E)

$^{16}\text{O}(p,pd)^{14}\text{N}$ 3.95 MeV (1^+ , $L=0$)

$^{16}\text{O}(p,nd)^{14}\text{O}$ 0.00 MeV (0^+ , $L=0$)

ratio: 18.1 ± 6.4 at 8.7°

16.9 ± 2.7 at 15.0°

→ strong suppression of *nn* pair

